

Trex SE

Application Info

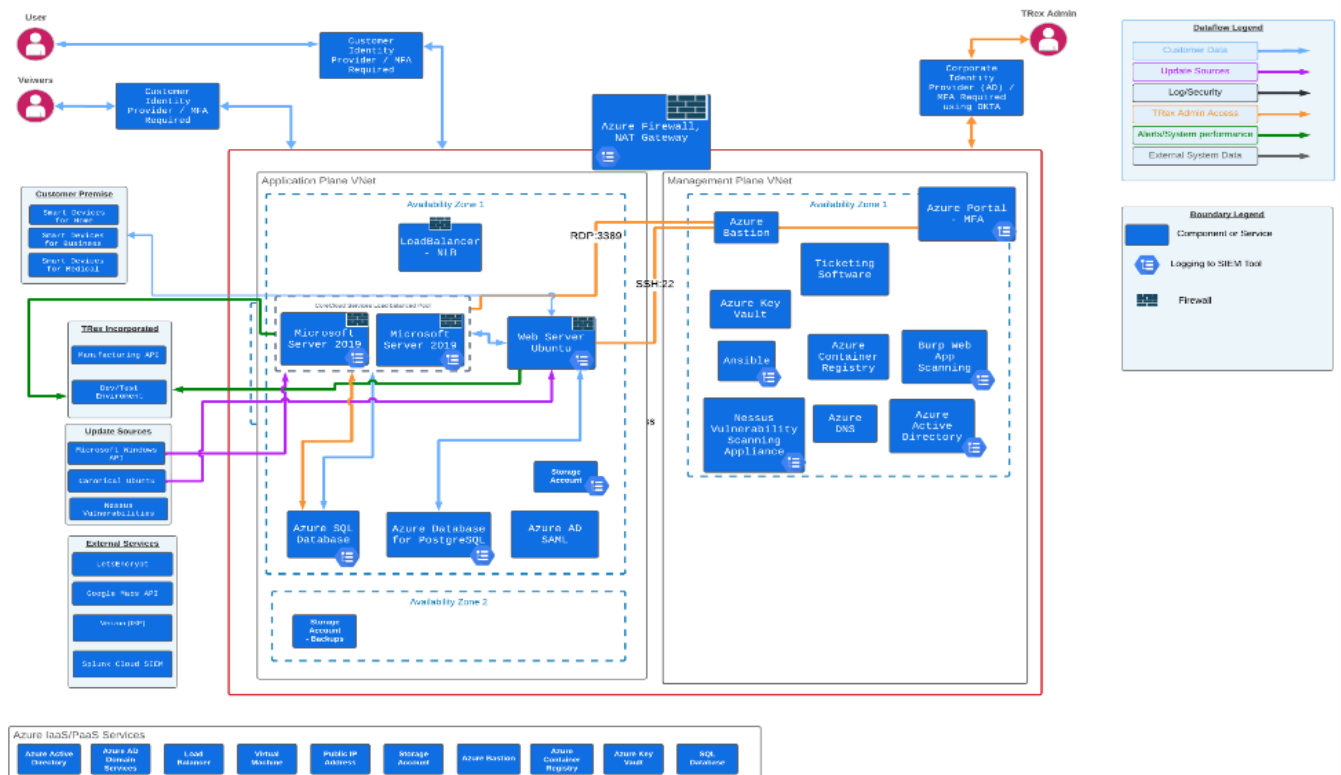
TRex, Inc. develops smart devices to be used for homes and businesses and in the medical industry. These smart devices include alarm systems, appliances, and human-worn devices. They also collect personal identifiable information (PII), protected health information (PHI), credit card information, device and security information, and other sensitive data. The devices can be configured to be linked to first responders, family members, or any authorized users that a business and/or the owner of the device chooses.

Architecture

Introduction

The devices that deploy TRex's software are configured to communicate with the TRex Smart Everything (TRex SE) Software as a Service (SaaS) platform that is hosted on the Azure Commercial Cloud Infrastructure as a Service (IaaS) platform. All TRex administrators log into the management plane of the TRex SE using TRex's active directory configured for multifactor authentication (MFA) using OKTA. Customers must use MFA and must provide their own identity provider (if applicable). All customer users will be configured to use MFA to gain access to their device as well as TRex SE, according to their choice, with no exceptions. FIPS 140-2 validated encryption is used throughout the environment, and all events from each host within the environment are logged and sent to Splunk. Within the TRex SE platform, all technology is updated automatically using advanced programmable interfaces (APIs). The home and business devices use their own internet service provider (ISP); however, due to the sensitive nature of the medical devices and their purposes, TRex will utilize Verizon ISP for internet connectivity with these devices. No data leaves the United States borders per service level agreements (SLA).

Architecture Diagram

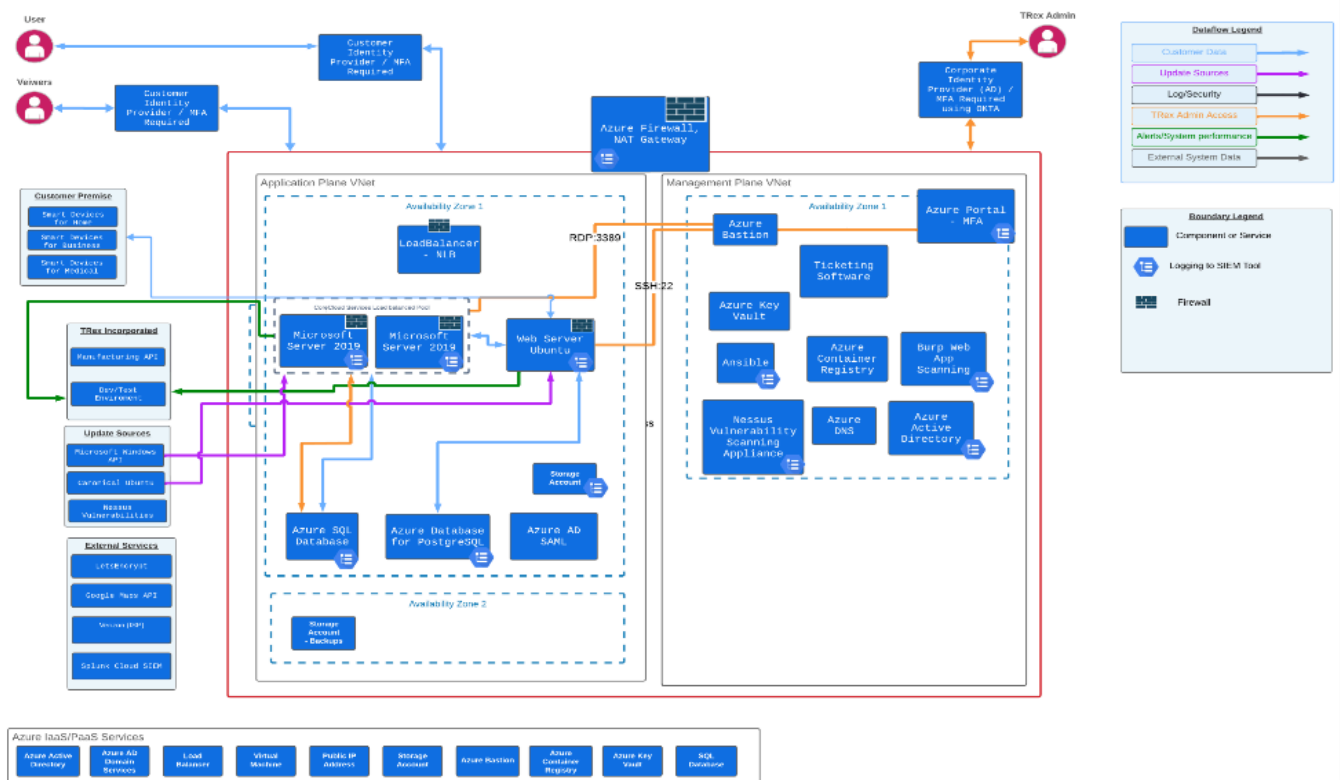


Dataflow

Introduction

There are several dataflows within the TRex SE environment including customer data, update sources, logging/security, TRex administration, alerts, and external system data. The two main dataflows are the customer to devices, devices to TRex SE, and management plane to application plane. Customers interact with their devices through an ISP provider. Although customers can access some offline functionality, there must be an active and stable internet connection for the device to be able to pull and push accurate and current information from TRex SE. Smart devices respond to customer voices, fingerprints, and in the case of medical devices, physical attributes such as heart rate, blood sugar levels, blood pressure, etc. This information, once collected from the customer, is communicated to TRex SE through an encrypted tunnel using FIPS 140-2 validated encryption. TRex SE analyzes the data sent from the device to either warn the user about potentially harmful activities or statistics, or if a certain range will prompt the user to dial 911 immediately, and for some medical devices, the device will use E911 and dial for the customer. For home and business devices, TRex SE may make recommendations based on the customer's use of the device as well as send health statistics of the device itself. For medical devices, TRex SE performs trend analysis to properly baseline the health of the individual using the device. This information is communicated to the user of the device but can also be communicated to authorized users, known as Viewers, of the information that the device produces. TRex admins are responsible for managing the TRex SE application. They perform several security and continuous monitoring activities.

Dataflow Diagram



Assumptions

Assumption Number	Assumption	Linked Threats	Linked Mitigations	Comments
A-0022	Trex Stores Customer PCI	T-0013 : A Trex Employee has no data backup schedule can cause a data loss crisis, which leads to permanent data	M-0005 : Trex Creates Firewall Rules to Block	

Assumption Number	Assumption	Linked Threats	Linked Mitigations	Comments
		loss, resulting in reduced availability and/or integrity of sensitive customer information, user account information, function of programs, network security and user account T-0012: A Trex employee has no firewall configuration review policy can increase network vulnerability, which leads to successful cyber attacks, resulting in reduced integrity and/or confidentiality of web server, AD Server, AD Database, sensitive customer information, user account information, network security, function of programs and network function T-0008: An internal actor Does not block incoming traffic to the Azure Firewall can allow external access to the network, which leads to degraded network security, resulting in reduced availability, integrity and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0007: An external threat actor bypasses azure firewall can access the network unknowingly, which leads to a network breach, resulting in reduced integrity, confidentiality and/or availability of web server, AD, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0006: An external threat actor Firewall rules are misconfigured, or there is an alternate route into the VNet that bypasses the Azure Firewall/NAT Gateway. can Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in the Application Plane, which leads to Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in the Application Plane, resulting in reduced availability, integrity and/or confidentiality of web	Inboud Traffic M-0002: Trex Requires MFA M-0008: Trex scans network on a schedule M-0009: Customers are required to sign into account every 2 weeks	

Assumption Number	Assumption	Linked Threats	Linked Mitigations	Comments
		server, AD, AD Server, AD Database and network function		
A-0021	Trex Inc locks admin accounts after 3 invalid login attempts	T-0009: A Trex Employee Does not take required cyber security training can leave employees vulnerable to cyber attacks, which leads to successful phishing attacks, resulting in reduced integrity, availability and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account information, network security, OKTA Token, function of programs and network function T-0010: A Trex Employee does not get locked out of account after 3 unsuccessful login attempts can an attacker can use brute force password attacks, which leads to unwarranted access to the network, resulting in reduced integrity and/or confidentiality of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token, network security and function of programs	M-0002: Trex Requires MFA M-0003: Trex Employees must take cyber security training	
A-0020	Trex Employees are trained on how to identify potential fishing attempts	T-0009: A Trex Employee Does not take required cyber security training can leave employees vulnerable to cyber attacks, which leads to successful phishing attacks, resulting in reduced integrity, availability and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account information, network security, OKTA Token, function of programs and network function T-0007: An external threat actor bypasses azure firewall can access the network unknowingly, which leads to a network breach, resulting in reduced integrity, confidentiality and/or availability of web server, AD, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0003: An external threat actor has external access to TCP port 22 can	M-0003: Trex Employees must take cyber security training M-0011: Trex runs phishing e-mail simulations	

Assumption Number	Assumption	Linked Threats	Linked Mitigations	Comments
		Brute force attack, which leads to remote access to the attacker, resulting in reduced integrity, confidentiality and/or availability of network function, user account, web server, AD Server and sensitive customer information		
A-0019	Update Sources and External Services are trusted and properly validated	T-0012: A Trex employee has no firewall configuration review policy can increase network vulnerability, which leads to successful cyber attacks, resulting in reduced integrity and/or confidentiality of web server, AD Server, AD Database, sensitive customer information, user account information, network security, function of programs and network function T-0007: An external threat actor bypasses azure firewall can access the network unknowingly, which leads to a network breach, resulting in reduced integrity, confidentiality and/or availability of web server, AD, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0002: An external threat actor traffic to port 53 is open can allow external access to the network, which leads to DDOS attacks are able to be performed, resulting in reduced integrity and/or availability of network function, web server, AD Database and AD Server T-0005: An internal actor does not keep SIEM programs up to date can cause potential network breaches to be missed, which leads to lower overall network security, resulting in reduced integrity of sensitive customer information, network function, user account, web server, AD, network security and function of programs T-0006: An external threat actor Firewall rules are misconfigured, or there is an alternate route into the VNet that bypasses the Azure Firewall/NAT Gateway. can Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in	M-0001: Trex maintains a quarterly patch management schedule M-0005: Trex Creates Firewall Rules to Block Inboud Traffic M-0008: Trex scans network on a schedule	

Assumption Number	Assumption	Linked Threats	Linked Mitigations	Comments
		the Application Plane, which leads to Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in the Application Plane, resulting in reduced availability, integrity and/or confidentiality of web server, AD, AD Server, AD Database and network function		
A-0018	Corporate Identity Provider (AD/Okta with MFA) governs all administrative access	T-0011: A Trex Employee never reviews user access rights can users having access to unnecessary systems, which leads to unauthorized configuration change, resulting in reduced integrity and/or availability of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token and function of programs T-0012: A Trex employee has no firewall configuration review policy can increase network vulnerability, which leads to successful cyber attacks, resulting in reduced integrity and/or confidentiality of web server, AD Server, AD Database, sensitive customer information, user account information, network security, function of programs and network function	M-0007 : Trex Reviews User Rights Quarterly	
A-0017	Trex Inc reviews admin access rights quarterly	T-0011: A Trex Employee never reviews user access rights can users having access to unnecessary systems, which leads to unauthorized configuration change, resulting in reduced integrity and/or availability of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token and function of programs	M-0007: Trex Reviews User Rights Quarterly M-0003: Trex Employees must take cyber security training	
A-0016	Trex Inc deploys a WAF (web application firewall)	T-0012: A Trex employee has no firewall configuration review policy can increase network vulnerability, which leads to successful cyber attacks, resulting in reduced integrity and/or confidentiality of web server, AD Server, AD Database, sensitive customer information, user account information, network security, function of programs and network function T-0007: An external threat actor bypasses azure firewall can access the	M-0005: Trex Creates Firewall Rules to Block Inbound Traffic M-0008: Trex scans network on a schedule	

Assumption Number	Assumption	Linked Threats	Linked Mitigations	Comments
		network unknowingly, which leads to a network breach, resulting in reduced integrity, confidentiality and/or availability of web server, AD, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0003: An external threat actor has external access to TCP port 22 can Brute force attack, which leads to remote access to the attacker, resulting in reduced integrity, confidentiality and/or availability of network function, user account, web server, AD Server and sensitive customer information T-0004: An external threat actor has access to TCP port 80 can access HTTP traffic in plain text, which leads to man in the middle attack, resulting in reduced confidentiality and/or integrity of sensitive customer information T-0006: An external threat actor Firewall rules are misconfigured, or there is an alternate route into the VNet that bypasses the Azure Firewall/NAT Gateway. can Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in the Application Plane, which leads to Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in the Application Plane, resulting in reduced availability, integrity and/or confidentiality of web server, AD, AD Server, AD Database and network function		
A-0015	Trex Inc Disables IP Address Forwarding	T-0012: A Trex employee has no firewall configuration review policy can increase network vulnerability, which leads to successful cyber attacks, resulting in reduced integrity and/or confidentiality of web server, AD Server, AD Database, sensitive customer information, user account information, network security, function of programs and network function T-0004: An external threat actor has access to TCP port 80 can access HTTP	M-0004: IP address forwarding is disabled for all OS M-0005: Trex Creates Firewall Rules to Block Inbound Traffic M-0008: Trex scans network on a schedule M-0010: Trex follows all vendor	

Assumption Number	Assumption	Linked Threats	Linked Mitigations	Comments
		traffic in plain text, which leads to man in the middle attack, resulting in reduced confidentiality and/or integrity of sensitive customer information	upgrade/end of life recommendations	
A-0014	Trex Inc runs scheduled updates on HTTP sever	T-0012: A Trex employee has no firewall configuration review policy can increase network vulnerability, which leads to successful cyber attacks, resulting in reduced integrity and/or confidentiality of web server, AD Server, AD Database, sensitive customer information, user account information, network security, function of programs and network function T-0002: An external threat actor traffic to port 53 is open can allow external access to the network, which leads to DDOS attacks are able to be performed, resulting in reduced integrity and/or availability of network function, web server, AD Database and AD Server T-0005: An internal actor does not keep SIEM programs up to date can cause potential network breaches to be missed, which leads to lower overall network security, resulting in reduced integrity of sensitive customer information, network function, user account, web server, AD, network security and function of programs	M-0001: Trex maintains a quarterly patch management schedule M-0005: Trex Creates Firewall Rules to Block Inboud Traffic M-0008: Trex scans network on a schedule	
A-0013	Customer Viewers Require MFA	T-0001: An internal actor Internal admin made an unauthorized change to the configuration of the active directory to not force MFA access for privileged users can bypass MFA, which leads to all users in AD permission would be changed, resulting in reduced confidentiality of AD, AD Server, AD Database and OKTA Token T-0010: A Trex Employee does not get locked out of account after 3 unsuccessful login attempts can an attacker can use brute force password attacks, which leads to unwarranted access to the network, resulting in reduced integrity and/or confidentiality of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token, network security and function of	M-0002: Trex Requires MFA M-0003: Trex Employees must take cyber security training M-0009: Customers are required to sign into account every 2 weeks	

Assumption Number	Assumption	Linked Threats	Linked Mitigations	Comments
		programs T-0007: An external threat actor bypasses azure firewall can access the network unknowingly, which leads to a network breach, resulting in reduced integrity, confidentiality and/or availability of web server, AD, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token		
A-0012	Only approved third party sources are trusted	T-0012: A Trex employee has no firewall configuration review policy can increase network vulnerability, which leads to successful cyber attacks, resulting in reduced integrity and/or confidentiality of web server, AD Server, AD Database, sensitive customer information, user account information, network security, function of programs and network function	M-0005: Trex Creates Firewall Rules to Block Inboud Traffic M-0008: Trex scans network on a schedule M-0010: Trex follows all vendor upgrade/end of life recommendations	
A-0011	Network only has one remote access service	T-0008: An internal actor Does not block incoming traffic to the Azure Firewall can allow external access to the network, which leads to degraded network security, resulting in reduced availability, integrity and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0012: A Trex employee has no firewall configuration review policy can increase network vulnerability, which leads to successful cyber attacks, resulting in reduced integrity and/or confidentiality of web server, AD Server, AD Database, sensitive customer information, user account information, network security, function of programs and network function T-0002: An external threat actor traffic to port 53 is open can allow external access to the network, which leads to DDOS attacks are able to be performed, resulting in reduced integrity and/or availability of network	M-0005: Trex Creates Firewall Rules to Block Inboud Traffic M-0008: Trex scans network on a schedule M-0010: Trex follows all vendor upgrade/end of life recommendations	

Assumption Number	Assumption	Linked Threats	Linked Mitigations	Comments
		function, web server, AD Database and AD Server T-0003 : An external threat actor has external access to TCP port 22 can Brute force attack, which leads to remote access to the attacker, resulting in reduced integrity, confidentiality and/or availability of network function, user account, web server, AD Server and sensitive customer information		
A-0010	Storage account backups are periodically updated	T-0013 : A Trex Employee has no data backup schedule can cause a data loss crisis, which leads to permanent data loss, resulting in reduced availability and/or integrity of sensitive customer information, user account information, function of programs, network security and user account	M-0006 : Trex Creates a data backup recovery plan	
A-0009	Application Plane vnet and Management Plane vnet are segmented	T-0012 : A Trex employee has no firewall configuration review policy can increase network vulnerability, which leads to successful cyber attacks, resulting in reduced integrity and/or confidentiality of web server, AD Server, AD Database, sensitive customer information, user account information, network security, function of programs and network function T-0003 : An external threat actor has external access to TCP port 22 can Brute force attack, which leads to remote access to the attacker, resulting in reduced integrity, confidentiality and/or availability of network function, user account, web server, AD Server and sensitive customer information	M-0005 : Trex Creates Firewall Rules to Block Inbound Traffic M-0008 : Trex scans network on a schedule	
A-0008	Alerts are sent to Trex Security for unusual server admin login attempts	T-0010 : A Trex Employee does not get locked out of account after 3 unsuccessful login attempts can an attacker can use brute force password attacks, which leads to unwarranted access to the network, resulting in reduced integrity and/or confidentiality of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token, network security and function of programs	M-0002 : Trex Requires MFA	
A-0007	Databases require admin	T-0011 : A Trex Employee never reviews user access rights can users having	M-0002 : Trex Requires MFA	

Assumption Number	Assumption	Linked Threats	Linked Mitigations	Comments
	privileged to access	access to unnecessary systems, which leads to unauthorized configuration change, resulting in reduced integrity and/or availability of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token and function of programs	M-0007 : Trex Reviews User Rights Quarterly	
A-0006	All incoming internet traffic must pass through Azure Firewall NAT Gateway	T-0006: An external threat actor Firewall rules are misconfigured, or there is an alternate route into the VNet that bypasses the Azure Firewall/NAT Gateway. can Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in the Application Plane, which leads to Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in the Application Plane, resulting in reduced availability, integrity and/or confidentiality of web server, AD, AD Server, AD Database and network function T-0002: An external threat actor traffic to port 53 is open can allow external access to the network, which leads to DDOS attacks are able to be performed, resulting in reduced integrity and/or availability of network function, web server, AD Database and AD Server T-0008: An internal actor Does not block incoming traffic to the Azure Firewall can allow external access to the network, which leads to degraded network security, resulting in reduced availability, integrity and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0003: An external threat actor has external access to TCP port 22 can Brute force attack, which leads to remote access to the attacker, resulting in reduced integrity, confidentiality and/or availability of network function, user account, web server, AD Server	M-0005: Trex Creates Firewall Rules to Block Inbound Traffic M-0008: Trex scans network on a schedule	

Assumption Number	Assumption	Linked Threats	Linked Mitigations	Comments
		and sensitive customer information T-0005 : An internal actor does not keep SIEM programs up to date can cause potential network breaches to be missed, which leads to lower overall network security, resulting in reduced integrity of sensitive customer information, network function, user account, web server, AD, network security and function of programs		
A-0005	Trex Inc keeps all SIEM programs updated on a regular schedule	T-0005: An internal actor does not keep SIEM programs up to date can cause potential network breaches to be missed, which leads to lower overall network security, resulting in reduced integrity of sensitive customer information, network function, user account, web server, AD, network security and function of programs T-0007: An external threat actor bypasses azure firewall can access the network unknowingly, which leads to a network breach, resulting in reduced integrity, confidentiality and/or availability of web server, AD, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0003: An external threat actor has external access to TCP port 22 can Brute force attack, which leads to remote access to the attacker, resulting in reduced integrity, confidentiality and/or availability of network function, user account, web server, AD Server and sensitive customer information T-0004: An external threat actor has access to TCP port 80 can access HTTP traffic in plain text, which leads to man in the middle attack, resulting in reduced confidentiality and/or integrity of sensitive customer information T-0006: An external threat actor Firewall rules are misconfigured, or there is an alternate route into the VNet that bypasses the Azure Firewall/NAT Gateway. can Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in	M-0001: Trex maintains a quarterly patch management schedule M-0008: Trex scans network on a schedule M-0010: Trex follows all vendor upgrade/end of life recommendations	

Assumption Number	Assumption	Linked Threats	Linked Mitigations	Comments
		the Application Plane, which leads to Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in the Application Plane, resulting in reduced availability, integrity and/or confidentiality of web server, AD, AD Server, AD Database and network function		
A-0004	Firewall closes TCP port 80	T-0004: An external threat actor has access to TCP port 80 can access HTTP traffic in plain text, which leads to man in the middle attack, resulting in reduced confidentiality and/or integrity of sensitive customer information T-0008: An internal actor Does not block incoming traffic to the Azure Firewall can allow external access to the network, which leads to degraded network security, resulting in reduced availability, integrity and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0012: A Trex employee has no firewall configuration review policy can increase network vulnerability, which leads to successful cyber attacks, resulting in reduced integrity and/or confidentiality of web server, AD Server, AD Database, sensitive customer information, user account information, network security, function of programs and network function T-0003: An external threat actor has external access to TCP port 22 can Brute force attack, which leads to remote access to the attacker, resulting in reduced integrity, confidentiality and/or availability of network function, user account, web server, AD Server and sensitive customer information T-0005: An internal actor does not keep SIEM programs up to date can cause potential network breaches to be missed, which leads to lower overall network security, resulting in reduced integrity of sensitive customer information, network function, user	M-0005: Trex Creates Firewall Rules to Block Inbound Traffic M-0008: Trex scans network on a schedule M-0010: Trex follows all vendor upgrade/end of life recommendations	

Assumption Number	Assumption	Linked Threats	Linked Mitigations	Comments
		account, web server, AD, network security and function of programs		
A-0003	Firewall closes TCP port 22	T-0003: An external threat actor has external access to TCP port 22 can Brute force attack, which leads to remote access to the attacker, resulting in reduced integrity, confidentiality and/or availability of network function, user account, web server, AD Server and sensitive customer information T-0008: An internal actor Does not block incoming traffic to the Azure Firewall can allow external access to the network, which leads to degraded network security, resulting in reduced availability, integrity and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0012: A Trex employee has no firewall configuration review policy can increase network vulnerability, which leads to successful cyber attacks, resulting in reduced integrity and/or confidentiality of web server, AD Server, AD Database, sensitive customer information, user account information, network security, function of programs and network function	M-0005: Trex Creates Firewall Rules to Block Inbound Traffic M-0008: Trex scans network on a schedule M-0010: Trex follows all vendor upgrade/end of life recommendations	
A-0002	Firewall closes port 53	T-0002: An external threat actor traffic to port 53 is open can allow external access to the network, which leads to DDOS attacks are able to be performed, resulting in reduced integrity and/or availability of network function, web server, AD Database and AD Server T-0008: An internal actor Does not block incoming traffic to the Azure Firewall can allow external access to the network, which leads to degraded network security, resulting in reduced availability, integrity and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and	M-0005: Trex Creates Firewall Rules to Block Inbound Traffic M-0008: Trex scans network on a schedule M-0010: Trex follows all vendor upgrade/end of life recommendations	

Assumption Number	Assumption	Linked Threats	Linked Mitigations	Comments
		OKTA Token T-0012 : A Trex employee has no firewall configuration review policy can increase network vulnerability, which leads to successful cyber attacks, resulting in reduced integrity and/or confidentiality of web server, AD Server, AD Database, sensitive customer information, user account information, network security, function of programs and network function		
A-0001	Trex Admin users must use MFA to access their accounts	T-0001 : An internal actor Internal admin made an unauthorized change to the configuration of the active directory to not force MFA access for privileged users can bypass MFA, which leads to all users in AD permission would be changed, resulting in reduced confidentiality of AD, AD Server, AD Database and OKTA Token T-0010 : A Trex Employee does not get locked out of account after 3 unsuccessful login attempts can an attacker can use brute force password attacks, which leads to unwarranted access to the network, resulting in reduced integrity and/or confidentiality of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token, network security and function of programs T-0007 : An external threat actor bypasses azure firewall can access the network unknowingly, which leads to a network breach, resulting in reduced integrity, confidentiality and/or availability of web server, AD, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token	M-0002 : Trex Requires MFA M-0007 : Trex Reviews User Rights Quarterly M-0003 : Trex Employees must take cyber security training	

Threats

Threat Number	Threat	Mitigations	Assumptions	Status	Priority	STRIDE	Comments
T-0013	A Trex Employee has no data	M-0001 : Trex maintains a quarterly patch	A-0010 : Storage account	Identified	High	R, I	

Threat Number	Threat	Mitigations	Assumptions	Status	Priority	STRIDE	Comments
	backup schedule can cause a data loss crisis, which leads to permanent data loss, resulting in reduced availability and/or integrity of sensitive customer information, user account information, function of programs, network security and user account	management schedule M-0006 : Trex Creates a data backup recovery plan	backups are periodically updated A-0022 : Trex Stores Customer PCI				
T-0012	A Trex employee has no firewall configuration review policy can increase network vulnerability, which leads to successful cyber attacks, resulting in reduced integrity and/or confidentiality of web server, AD Server, AD Database, sensitive customer information, user account information, network security, function of programs and network function	M-0005 : Trex Creates Firewall Rules to Block Inboud Traffic M-0008 : Trex scans network on a schedule M-0010 : Trex follows all vendor upgrade/end of life recommendations	A-0019 : Update Sources and External Services are trusted and properly validated A-0016 : Trex Inc deploys a WAF (web application firewall) A-0015 : Trex Inc Disables IP Address Forwarding A-0014 : Trex Inc runs scheduled updates on HTTP sever A-0012 : Only approved third party sources are trusted A-0009 : Application Plane vnet	Identified	High	T, R, D	

Threat Number	Threat	Mitigations	Assumptions	Status	Priority	STRIDE	Comments
			and Management Plane vnet are segmented A-0004: Firewall closes TCP port 80 A-0003: Firewall closes TCP port 22 A-0002: Firewall closes port 53 A-0011: Network only has one remote access service A-0018: Corporate Identity Provider (AD/Okta with MFA) governs all administrative access A-0022: Trex Stores Customer PCI				
T-0011	A Trex Employee never reviews user access rights can users having access to unnecessary systems, which leads to unauthorized configuration change, resulting in reduced integrity and/or availability of AD Server, web server,	M-0007: Trex Reviews User Rights Quarterly M-0003: Trex Employees must take cyber security training	A-0018: Corporate Identity Provider (AD/Okta with MFA) governs all administrative access A-0017: Trex Inc reviews admin access rights quarterly A-0007: Databases require admin privileged to access	Identified	Low	I, E, R	

Threat Number	Threat	Mitigations	Assumptions	Status	Priority	STRIDE	Comments
	AD Database, sensitive customer information, user account information, OKTA Token and function of programs						
T-0010	A Trex Employee does not get locked out of account after 3 unsuccessful login attempts can an attacker can use brute force password attacks, which leads to unwarranted access to the network, resulting in reduced integrity and/or confidentiality of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token, network security and function of programs	M-0002: Trex Requires MFA M-0003: Trex Employees must take cyber security training	A-0021: Trex Inc locks admin accounts after 3 invalid login attempts A-0013: Customer Viewers Require MFA A-0008: Alerts are sent to Trex Security for unusual server admin login attempts A-0001: Trex Admin users must use MFA to access their accounts	Identified	Medium	S, R, I, E, T	
T-0009	A Trex Employee Does not take required cyber security training can leave	M-0003: Trex Employees must take cyber security training M-0011: Trex runs phishing e-mail simulations	A-0021: Trex Inc locks admin accounts after 3 invalid login attempts A-0020: Trex	Identified	Medium	T, R, I, E, S	

Threat Number	Threat	Mitigations	Assumptions	Status	Priority	STRIDE	Comments
	employees vulnerable to cyber attacks, which leads to successful phishing attacks, resulting in reduced integrity, availability and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account information, network security, OKTA Token, function of programs and network function		Employees are trained on how to identify potential fishing attempts				
T-0008	An internal actor Does not block incoming traffic to the Azure Firewall can allow external access to the network, which leads to degraded network security, resulting in reduced availability, integrity and/or confidentiality of AD, web server, AD Server, AD	M-0004 : IP address forwarding is disabled for all OS M-0005 : Trex Creates Firewall Rules to Block Inboud Traffic M-0008 : Trex scans network on a schedule M-0010 : Trex follows all vendor upgrade/end of life recommendations	A-0002 : Firewall closes port 53 A-0003 : Firewall closes TCP port 22 A-0004 : Firewall closes TCP port 80 A-0006 : All incoming internet traffic must pass through Azure Firewall NAT Gateway A-0011 : Network only has one remote access service	Identified	Medium	R, I, E	

Threat Number	Threat	Mitigations	Assumptions	Status	Priority	STRIDE	Comments
	Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token		A-0022 : Trex Stores Customer PCI				
T-0007	An external threat actor bypasses azure firewall can access the network unknowingly, which leads to a network breach, resulting in reduced integrity, confidentiality and/or availability of web server, AD, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token	M-0005 : Trex Creates Firewall Rules to Block Inboud Traffic M-0008 : Trex scans network on a schedule M-0010 : Trex follows all vendor upgrade/end of life recommendations	A-0001 : Trex Admin users must use MFA to access their accounts A-0013 : Customer Viewers Require MFA A-0005 : Trex Inc keeps all SIEM programs updated on a regular schedule A-0016 : Trex Inc deploys a WAF (web application firewall) A-0019 : Update Sources and External Services are trusted and properly validated A-0020 : Trex Employees are trained on how to identify potential fishing attempts A-0022 : Trex Stores Customer PCI	Identified	High	T, R, I, E	

Threat Number	Threat	Mitigations	Assumptions	Status	Priority	STRIDE	Comments
T-0006	An external threat actor Firewall rules are misconfigured, or there is an alternate route into the VNet that bypasses the Azure Firewall/NAT Gateway. can Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in the Application Plane, which leads to Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in the Application Plane, resulting in reduced availability, integrity and/or confidentiality of web server, AD, AD Server, AD Database and network function	M-0005 : Trex Creates Firewall Rules to Block Inbound Traffic M-0008 : Trex scans network on a schedule M-0010 : Trex follows all vendor upgrade/end of life recommendations	A-0006 : All incoming internet traffic must pass through Azure Firewall NAT Gateway A-0019 : Update Sources and External Services are trusted and properly validated A-0005 : Trex Inc keeps all SIEM programs updated on a regular schedule A-0016 : Trex Inc deploys a WAF (web application firewall) A-0022 : Trex Stores Customer PCI	Identified	High	T, I, E	

Threat Number	Threat	Mitigations	Assumptions	Status	Priority	STRIDE	Comments
T-0005	An internal actor does not keep SIEM programs up to date can cause potential network breaches to be missed, which leads to lower overall network security, resulting in reduced integrity of sensitive customer information, network function, user account, web server, AD, network security and function of programs	M-0001: Trex maintains a quarterly patch management schedule M-0008: Trex scans network on a schedule M-0010: Trex follows all vendor upgrade/end of life recommendations	A-0005: Trex Inc keeps all SIEM programs updated on a regular schedule A-0004: Firewall closes TCP port 80 A-0006: All incoming internet traffic must pass through Azure Firewall NAT Gateway A-0019: Update Sources and External Services are trusted and properly validated A-0014: Trex Inc runs scheduled updates on HTTP sever	Identified	Medium	R	
T-0004	An external threat actor has access to TCP port 80 can access HTTP traffic in plain text, which leads to man in the middle attack, resulting in reduced confidentiality and/or integrity of sensitive customer information	M-0005: Trex Creates Firewall Rules to Block Inboud Traffic M-0008: Trex scans network on a schedule M-0010: Trex follows all vendor upgrade/end of life recommendations	A-0004: Firewall closes TCP port 80 A-0005: Trex Inc keeps all SIEM programs updated on a regular schedule A-0016: Trex Inc deploys a WAF (web application firewall) A-0015: Trex Inc Disables IP Address Forwarding	Identified	High	T, R, I	

Threat Number	Threat	Mitigations	Assumptions	Status	Priority	STRIDE	Comments
T-0003	An external threat actor has external access to TCP port 22 can Brute force attack, which leads to remote access to the attacker, resulting in reduced integrity, confidentiality and/or availability of network function, user account, web server, AD Server and sensitive customer information	M-0005: Trex Creates Firewall Rules to Block Inboud Traffic M-0008: Trex scans network on a schedule M-0009: Customers are required to sign into account every 2 weeks M-0010: Trex follows all vendor upgrade/end of life recommendations	A-0003: Firewall closes TCP port 22 A-0020: Trex Employees are trained on how to identify potential fishing attempts A-0016: Trex Inc deploys a WAF (web application firewall) A-0011: Network only has one remote access service A-0009: Application Plane vnet and Management Plane vnet are segmented A-0006: All incoming internet traffic must pass through Azure Firewall NAT Gateway A-0004: Firewall closes TCP port 80 A-0005: Trex Inc keeps all SIEM programs updated on a regular schedule	Identified	High	T, I, E, S	
T-0002	An external threat actor traffic to port 53 is open can allow external	M-0005: Trex Creates Firewall Rules to Block Inboud Traffic M-0008: Trex	A-0002: Firewall closes port 53 A-0006: All incoming	Identified	High	D	

Threat Number	Threat	Mitigations	Assumptions	Status	Priority	STRIDE	Comments
	access to the network, which leads to DDOS attacks are able to be performed, resulting in reduced integrity and/or availability of network function, web server, AD Database and AD Server	scans network on a schedule M-0010 : Trex follows all vendor upgrade/end of life recommendations	internet traffic must pass through Azure Firewall NAT Gateway A-0011 : Network only has one remote access service A-0019 : Update Sources and External Services are trusted and properly validated A-0014 : Trex Inc runs scheduled updates on HTTP sever				
T-0001	An internal actor Internal admin made an unauthorized change to the configuration of the active directory to not force MFA access for privileged users can bypass MFA, which leads to all users in AD permission would be changed, resulting in reduced confidentiality of AD, AD Server, AD Database and OKTA Token	M-0002 : Trex Requires MFA M-0007 : Trex Reviews User Rights Quarterly M-0003 : Trex Employees must take cyber security training M-0010 : Trex follows all vendor upgrade/end of life recommendations	A-0001 : Trex Admin users must use MFA to access their accounts A-0013 : Customer Viewers Require MFA	Identified	High	I	

Mitigations

Mitigation Number	Mitigation	Threats Mitigating	Assumptions	Status	Comments
M-0011	Trex runs phishing e-mail simulations	T-0009 : A Trex Employee Does not take required cyber security training can leave employees vulnerable to cyber attacks, which leads to successful phishing attacks, resulting in reduced integrity, availability and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account information, network security, OKTA Token, function of programs and network function	A-0020 : Trex Employees are trained on how to identify potential fishing attempts	Identified	
M-0010	Trex follows all vendor upgrade/end of life recommendations	T-0008: An internal actor Does not block incoming traffic to the Azure Firewall can allow external access to the network, which leads to degraded network security, resulting in reduced availability, integrity and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0007: An external threat actor bypasses azure firewall can access the network unknowingly, which leads to a network breach, resulting in reduced integrity, confidentiality and/or availability of web server, AD, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0006: An external threat actor Firewall rules are misconfigured, or there is an alternate route into the VNet that bypasses the Azure Firewall/NAT Gateway. can Attackers discover and use	A-0002: Firewall closes port 53 A-0004: Firewall closes TCP port 80 A-0003: Firewall closes TCP port 22 A-0005: Trex Inc keeps all SIEM programs updated on a regular schedule A-0011: Network only has one remote access service A-0012: Only approved third party sources are trusted A-0015: Trex Inc Disables IP Address Forwarding	Identified	

Mitigation Number	Mitigation	Threats Mitigating	Assumptions	Status	Comments
		the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in the Application Plane, which leads to Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in the Application Plane, resulting in reduced availability, integrity and/or confidentiality of web server, AD, AD Server, AD Database and network function T-0005: An internal actor does not keep SIEM programs up to date can cause potential network breaches to be missed, which leads to lower overall network security, resulting in reduced integrity of sensitive customer information, network function, user account, web server, AD, network security and function of programs T-0004: An external threat actor has access to TCP port 80 can access HTTP traffic in plain text, which leads to man in the middle attack, resulting in reduced confidentiality and/or integrity of sensitive customer information T-0003: An external threat actor has external access to TCP port 22 can Brute force attack, which leads to remote access to the attacker, resulting in reduced integrity, confidentiality and/or availability of network function, user account, web server, AD Server and sensitive customer information T-0012: A Trex employee has no firewall configuration review policy can increase network vulnerability, which leads to successful cyber			

Mitigation Number	Mitigation	Threats Mitigating	Assumptions	Status	Comments
		attacks, resulting in reduced integrity and/or confidentiality of web server, AD Server, AD Database, sensitive customer information, user account information, network security, function of programs and network function T-0002: An external threat actor traffic to port 53 is open can allow external access to the network, which leads to DDOS attacks are able to be performed, resulting in reduced integrity and/or availability of network function, web server, AD Database and AD Server T-0001: An internal actor Internal admin made an unauthorized change to the configuration of the active directory to not force MFA access for privileged users can bypass MFA, which leads to all users in AD permission would be changed, resulting in reduced confidentiality of AD, AD Server, AD Database and OKTA Token			
M-0009	Customers are required to sign into account every 2 weeks	T-0003: An external threat actor has external access to TCP port 22 can Brute force attack, which leads to remote access to the attacker, resulting in reduced integrity, confidentiality and/or availability of network function, user account, web server, AD Server and sensitive customer information	A-0022: Trex Stores Customer PCI A-0013: Customer Viewers Require MFA	Identified	
M-0008	Trex scans network on a schedule	T-0012: A Trex employee has no firewall configuration review policy can increase network vulnerability, which leads to successful cyber attacks, resulting in reduced integrity and/or	A-0022: Trex Stores Customer PCI A-0019: Update Sources and External Services are	Identified	

Mitigation Number	Mitigation	Threats Mitigating	Assumptions	Status	Comments
		confidentiality of web server, AD Server, AD Database, sensitive customer information, user account information, network security, function of programs and network function T-0008: An internal actor Does not block incoming traffic to the Azure Firewall can allow external access to the network, which leads to degraded network security, resulting in reduced availability, integrity and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0007: An external threat actor bypasses azure firewall can access the network unknowingly, which leads to a network breach, resulting in reduced integrity, confidentiality and/or availability of web server, AD, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0006: An external threat actor Firewall rules are misconfigured, or there is an alternate route into the VNet that bypasses the Azure Firewall/NAT Gateway. can Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in the Application Plane, which leads to Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly	trusted and properly validated A-0016: Trex Inc deploys a WAF (web application firewall) A-0015: Trex Inc Disables IP Address Forwarding A-0014: Trex Inc runs scheduled updates on HTTP sever A-0012: Only approved third party sources are trusted A-0011: Network only has one remote access service A-0009: Application Plane vnet and Management Plane vnet are segmented A-0006: All incoming internet traffic must pass through Azure Firewall NAT Gateway A-0005: Trex Inc keeps all SIEM programs updated on a regular schedule A-0004: Firewall closes TCP port 80 A-0003: Firewall closes TCP port 22 A-0002: Firewall closes port 53		

Mitigation Number	Mitigation	Threats Mitigating	Assumptions	Status	Comments
		to servers in the Application Plane, resulting in reduced availability, integrity and/or confidentiality of web server, AD, AD Server, AD Database and network function T-0005: An internal actor does not keep SIEM programs up to date can cause potential network breaches to be missed, which leads to lower overall network security, resulting in reduced integrity of sensitive customer information, network function, user account, web server, AD, network security and function of programs T-0004: An external threat actor has access to TCP port 80 can access HTTP traffic in plain text, which leads to man in the middle attack, resulting in reduced confidentiality and/or integrity of sensitive customer information T-0003: An external threat actor has external access to TCP port 22 can Brute force attack, which leads to remote access to the attacker, resulting in reduced integrity, confidentiality and/or availability of network function, user account, web server, AD Server and sensitive customer information T-0002: An external threat actor traffic to port 53 is open can allow external access to the network, which leads to DDOS attacks are able to be performed, resulting in reduced integrity and/or availability of network function, web server, AD Database and AD Server			
M-0007	Trex Reviews User Rights Quarterly	T-0011 : A Trex Employee never reviews user access rights can users having access	A-0018 : Corporate Identity	Identified	

Mitigation Number	Mitigation	Threats Mitigating	Assumptions	Status	Comments
		to unnecessary systems, which leads to unauthorized configuration change, resulting in reduced integrity and/or availability of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token and function of programs T-0001 : An internal actor Internal admin made an unauthorized change to the configuration of the active directory to not force MFA access for privileged users can bypass MFA, which leads to all users in AD permission would be changed, resulting in reduced confidentiality of AD, AD Server, AD Database and OKTA Token	Provider (AD/Okta with MFA) governs all administrative access A-0017 : Trex Inc reviews admin access rights quarterly A-0007 : Databases require admin privileged to access A-0001 : Trex Admin users must use MFA to access their accounts		
M-0006	Trex Creates a data backup recovery plan	T-0013 : A Trex Employee has no data backup schedule can cause a data loss crisis, which leads to permanent data loss, resulting in reduced availability and/or integrity of sensitive customer information, user account information, function of programs, network security and user account	A-0010 : Storage account backups are periodically updated	Identified	
M-0005	Trex Creates Firewall Rules to Block Inbound Traffic	T-0012 : A Trex employee has no firewall configuration review policy can increase network vulnerability, which leads to successful cyber attacks, resulting in reduced integrity and/or confidentiality of web server, AD Server, AD Database, sensitive customer information, user account information, network security, function of programs and network function T-0008 : An internal actor Does not block incoming traffic to the Azure Firewall	A-0019 : Update Sources and External Services are trusted and properly validated A-0016 : Trex Inc deploys a WAF (web application firewall) A-0014 : Trex Inc runs scheduled updates on HTTP sever A-0015 : Trex Inc Disables IP	Identified	

Mitigation Number	Mitigation	Threats Mitigating	Assumptions	Status	Comments
		can allow external access to the network, which leads to degraded network security, resulting in reduced availability, integrity and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0007: An external threat actor bypasses azure firewall can access the network unknowingly, which leads to a network breach, resulting in reduced integrity, confidentiality and/or availability of web server, AD, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0006: An external threat actor Firewall rules are misconfigured, or there is an alternate route into the VNet that bypasses the Azure Firewall/NAT Gateway. can Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in the Application Plane, which leads to Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in the Application Plane, resulting in reduced availability, integrity and/or confidentiality of web server, AD, AD Server, AD Database and network function T-0004: An external threat actor has access to TCP port 80 can access HTTP traffic in plain text, which leads to man in the middle attack, resulting	Address Forwarding A-0011: Network only has one remote access service A-0012: Only approved third party sources are trusted A-0009: Application Plane vnet and Management Plane vnet are segmented A-0006: All incoming internet traffic must pass through Azure Firewall NAT Gateway A-0004: Firewall closes TCP port 80 A-0003: Firewall closes TCP port 22 A-0002: Firewall closes port 53 A-0022: Trex Stores Customer PCI		

Mitigation Number	Mitigation	Threats Mitigating	Assumptions	Status	Comments
		in reduced confidentiality and/or integrity of sensitive customer information T-0003: An external threat actor has external access to TCP port 22 can Brute force attack, which leads to remote access to the attacker, resulting in reduced integrity, confidentiality and/or availability of network function, user account, web server, AD Server and sensitive customer information T-0002: An external threat actor traffic to port 53 is open can allow external access to the network, which leads to DDOS attacks are able to be performed, resulting in reduced integrity and/or availability of network function, web server, AD Database and AD Server			
M-0004	IP address forwarding is disabled for all OS	T-0008: An internal actor Does not block incoming traffic to the Azure Firewall can allow external access to the network, which leads to degraded network security, resulting in reduced availability, integrity and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token	A-0015 : Trex Inc Disables IP Address Forwarding	Identified	
M-0003	Trex Employees must take cyber security training	T-0009: A Trex Employee Does not take required cyber security training can leave employees vulnerable to cyber attacks, which leads to successful phishing attacks, resulting in reduced integrity, availability and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer	A-0020: Trex Employees are trained on how to identify potential fishing attempts A-0001: Trex Admin users must use MFA to access their accounts	Identified	

Mitigation Number	Mitigation	Threats Mitigating	Assumptions	Status	Comments
		information, user account information, network security, OKTA Token, function of programs and network function T-0010: A Trex Employee does not get locked out of account after 3 unsuccessful login attempts can an attacker can use brute force password attacks, which leads to unwarranted access to the network, resulting in reduced integrity and/or confidentiality of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token, network security and function of programs T-0011: A Trex Employee never reviews user access rights can users having access to unnecessary systems, which leads to unauthorized configuration change, resulting in reduced integrity and/or availability of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token and function of programs T-0001: An internal actor Internal admin made an unauthorized change to the configuration of the active directory to not force MFA access for privileged users can bypass MFA, which leads to all users in AD permission would be changed, resulting in reduced confidentiality of AD, AD Server, AD Database and OKTA Token	A-0021: Trex Inc locks admin accounts after 3 invalid login attempts A-0017: Trex Inc reviews admin access rights quarterly A-0013: Customer Viewers Require MFA		
M-0002	Trex Requires MFA	T-0010: A Trex Employee does not get locked out of account after 3 unsuccessful login attempts can an attacker can use brute force password attacks, which	A-0021: Trex Inc locks admin accounts after 3 invalid login attempts A-0013:	Identified	

Mitigation Number	Mitigation	Threats Mitigating	Assumptions	Status	Comments
		leads to unwarranted access to the network, resulting in reduced integrity and/or confidentiality of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token, network security and function of programs T-0001: An internal actor Internal admin made an unauthorized change to the configuration of the active directory to not force MFA access for privileged users can bypass MFA, which leads to all users in AD permission would be changed, resulting in reduced confidentiality of AD, AD Server, AD Database and OKTA Token	Customer Viewers Require MFA A-0008: Alerts are sent to Trex Security for unusual server admin login attempts A-0007: Databases require admin privileged to access A-0001: Trex Admin users must use MFA to access their accounts A-0022: Trex Stores Customer PCI		
M-0001	Trex maintains a quarterly patch management schedule	T-0013: A Trex Employee has no data backup schedule can cause a data loss crisis, which leads to permanent data loss, resulting in reduced availability and/or integrity of sensitive customer information, user account information, function of programs, network security and user account T-0005: An internal actor does not keep SIEM programs up to date can cause potential network breaches to be missed, which leads to lower overall network security, resulting in reduced integrity of sensitive customer information, network function, user account, web server, AD, network security and function of programs	A-0019: Update Sources and External Services are trusted and properly validated A-0014: Trex Inc runs scheduled updates on HTTP sever A-0005: Trex Inc keeps all SIEM programs updated on a regular schedule	Identified	

Impacted Assets

Assets Number	Asset	Related Threats
AS-0001	sensitive customer information	T-0013: A Trex Employee has no data backup schedule can cause a data loss crisis, which leads to permanent data loss, resulting in reduced availability and/or integrity of sensitive customer information, user account information, function of programs, network security and user account T-0012: A Trex employee has no firewall configuration review policy can increase network vulnerability, which leads to successful cyber attacks, resulting in reduced integrity and/or confidentiality of web server, AD Server, AD Database, sensitive customer information, user account information, network security, function of programs and network function T-0011: A Trex Employee never reviews user access rights can users having access to unnecessary systems, which leads to unauthorized configuration change, resulting in reduced integrity and/or availability of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token and function of programs T-0010: A Trex Employee does not get locked out of account after 3 unsuccessful login attempts can an attacker can use brute force password attacks, which leads to unwarranted access to the network, resulting in reduced integrity and/or confidentiality of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token, network security and function of programs T-0009: A Trex Employee Does not take required cyber security training can leave employees vulnerable to cyber attacks, which leads to successful phishing attacks, resulting in reduced integrity, availability and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account information, network security, OKTA Token, function of programs and network function T-0008: An internal actor Does not block incoming traffic to the Azure Firewall can allow external access to the network, which leads to degraded network security, resulting in reduced availability, integrity and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0007: An external threat actor bypasses azure firewall can access the network unknowingly, which leads to a network breach, resulting in reduced integrity, confidentiality and/or availability of web server, AD, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0005: An internal actor does not keep SIEM programs up to date can cause potential network breaches to be missed, which leads to lower overall network security, resulting in reduced integrity of sensitive customer information, network function, user account, web server, AD, network security and function of programs T-0004: An external threat actor has access to TCP port 80 can access HTTP traffic in plain text, which leads to man in the middle attack, resulting in reduced confidentiality and/or integrity of sensitive customer information T-0003: An external threat actor has external access to TCP port 22 can Brute force attack, which leads to remote access to the attacker, resulting in reduced integrity, confidentiality and/or availability of network function, user account, web server, AD Server and sensitive customer information
AS-0002	user account information	T-0013: A Trex Employee has no data backup schedule can cause a data loss crisis, which leads to permanent data loss, resulting in reduced availability and/or integrity of sensitive customer information, user account information, function of programs, network security and user account T-0012: A Trex employee has no firewall configuration review policy can increase network vulnerability, which leads to successful cyber attacks, resulting in reduced integrity and/or confidentiality of web server, AD Server, AD Database, sensitive

Assets Number	Asset	Related Threats
		customer information, user account information, network security, function of programs and network function T-0011: A Trex Employee never reviews user access rights can users having access to unnecessary systems, which leads to unauthorized configuration change, resulting in reduced integrity and/or availability of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token and function of programs T-0010: A Trex Employee does not get locked out of account after 3 unsuccessful login attempts can an attacker can use brute force password attacks, which leads to unwarranted access to the network, resulting in reduced integrity and/or confidentiality of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token, network security and function of programs T-0009: A Trex Employee Does not take required cyber security training can leave employees vulnerable to cyber attacks, which leads to successful phishing attacks, resulting in reduced integrity, availability and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account information, network security, OKTA Token, function of programs and network function T-0008: An internal actor Does not block incoming traffic to the Azure Firewall can allow external access to the network, which leads to degraded network security, resulting in reduced availability, integrity and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0007: An external threat actor bypasses azure firewall can access the network unknowingly, which leads to a network breach, resulting in reduced integrity, confidentiality and/or availability of web server, AD, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token
AS-0003	function of programs	T-0013: A Trex Employee has no data backup schedule can cause a data loss crisis, which leads to permanent data loss, resulting in reduced availability and/or integrity of sensitive customer information, user account information, function of programs, network security and user account T-0012: A Trex employee has no firewall configuration review policy can increase network vulnerability, which leads to successful cyber attacks, resulting in reduced integrity and/or confidentiality of web server, AD Server, AD Database, sensitive customer information, user account information, network security, function of programs and network function T-0011: A Trex Employee never reviews user access rights can users having access to unnecessary systems, which leads to unauthorized configuration change, resulting in reduced integrity and/or availability of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token and function of programs T-0010: A Trex Employee does not get locked out of account after 3 unsuccessful login attempts can an attacker can use brute force password attacks, which leads to unwarranted access to the network, resulting in reduced integrity and/or confidentiality of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token, network security and function of programs T-0009: A Trex Employee Does not take required cyber security training can leave employees vulnerable to cyber attacks, which leads to successful phishing attacks, resulting in reduced integrity, availability and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account information, network security, OKTA Token, function of programs and network function T-0008: An internal actor Does not block incoming traffic to the Azure Firewall can

Assets Number	Asset	Related Threats
		allow external access to the network, which leads to degraded network security, resulting in reduced availability, integrity and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0007: An external threat actor bypasses azure firewall can access the network unknowingly, which leads to a network breach, resulting in reduced integrity, confidentiality and/or availability of web server, AD, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0005: An internal actor does not keep SIEM programs up to date can cause potential network breaches to be missed, which leads to lower overall network security, resulting in reduced integrity of sensitive customer information, network function, user account, web server, AD, network security and function of programs
AS-0004	network security	T-0013: A Trex Employee has no data backup schedule can cause a data loss crisis, which leads to permanent data loss, resulting in reduced availability and/or integrity of sensitive customer information, user account information, function of programs, network security and user account T-0012: A Trex employee has no firewall configuration review policy can increase network vulnerability, which leads to successful cyber attacks, resulting in reduced integrity and/or confidentiality of web server, AD Server, AD Database, sensitive customer information, user account information, network security, function of programs and network function T-0010: A Trex Employee does not get locked out of account after 3 unsuccessful login attempts can an attacker can use brute force password attacks, which leads to unwarranted access to the network, resulting in reduced integrity and/or confidentiality of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token, network security and function of programs T-0009: A Trex Employee Does not take required cyber security training can leave employees vulnerable to cyber attacks, which leads to successful phishing attacks, resulting in reduced integrity, availability and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account information, network security, OKTA Token, function of programs and network function T-0008: An internal actor Does not block incoming traffic to the Azure Firewall can allow external access to the network, which leads to degraded network security, resulting in reduced availability, integrity and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0007: An external threat actor bypasses azure firewall can access the network unknowingly, which leads to a network breach, resulting in reduced integrity, confidentiality and/or availability of web server, AD, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0005: An internal actor does not keep SIEM programs up to date can cause potential network breaches to be missed, which leads to lower overall network security, resulting in reduced integrity of sensitive customer information, network function, user account, web server, AD, network security and function of programs
AS-0005	user account	T-0013: A Trex Employee has no data backup schedule can cause a data loss crisis, which leads to permanent data loss, resulting in reduced availability and/or integrity of sensitive customer information, user account information, function of programs, network security and user account T-0008: An internal actor Does not block incoming traffic to the Azure Firewall can

Assets Number	Asset	Related Threats
		allow external access to the network, which leads to degraded network security, resulting in reduced availability, integrity and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0007: An external threat actor bypasses azure firewall can access the network unknowingly, which leads to a network breach, resulting in reduced integrity, confidentiality and/or availability of web server, AD, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0005: An internal actor does not keep SIEM programs up to date can cause potential network breaches to be missed, which leads to lower overall network security, resulting in reduced integrity of sensitive customer information, network function, user account, web server, AD, network security and function of programs T-0003: An external threat actor has external access to TCP port 22 can Brute force attack, which leads to remote access to the attacker, resulting in reduced integrity, confidentiality and/or availability of network function, user account, web server, AD Server and sensitive customer information
AS-0006	web server	T-0012: A Trex employee has no firewall configuration review policy can increase network vulnerability, which leads to successful cyber attacks, resulting in reduced integrity and/or confidentiality of web server, AD Server, AD Database, sensitive customer information, user account information, network security, function of programs and network function T-0011: A Trex Employee never reviews user access rights can users having access to unnecessary systems, which leads to unauthorized configuration change, resulting in reduced integrity and/or availability of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token and function of programs T-0010: A Trex Employee does not get locked out of account after 3 unsuccessful login attempts can an attacker can use brute force password attacks, which leads to unwarranted access to the network, resulting in reduced integrity and/or confidentiality of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token, network security and function of programs T-0009: A Trex Employee Does not take required cyber security training can leave employees vulnerable to cyber attacks, which leads to successful phishing attacks, resulting in reduced integrity, availability and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account information, network security, OKTA Token, function of programs and network function T-0008: An internal actor Does not block incoming traffic to the Azure Firewall can allow external access to the network, which leads to degraded network security, resulting in reduced availability, integrity and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0007: An external threat actor bypasses azure firewall can access the network unknowingly, which leads to a network breach, resulting in reduced integrity, confidentiality and/or availability of web server, AD, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0006: An external threat actor Firewall rules are misconfigured, or there is an alternate route into the VNet that bypasses the Azure Firewall/NAT Gateway. can Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in the Application Plane, which leads to Attackers discover and use the unprotected path (bypassing the firewall) to send malicious

Assets Number	Asset	Related Threats
		traffic directly to servers in the Application Plane, resulting in reduced availability, integrity and/or confidentiality of web server, AD, AD Server, AD Database and network function T-0005: An internal actor does not keep SIEM programs up to date can cause potential network breaches to be missed, which leads to lower overall network security, resulting in reduced integrity of sensitive customer information, network function, user account, web server, AD, network security and function of programs T-0003: An external threat actor has external access to TCP port 22 can Brute force attack, which leads to remote access to the attacker, resulting in reduced integrity, confidentiality and/or availability of network function, user account, web server, AD Server and sensitive customer information T-0002: An external threat actor traffic to port 53 is open can allow external access to the network, which leads to DDOS attacks are able to be performed, resulting in reduced integrity and/or availability of network function, web server, AD Database and AD Server
AS-0007	AD Server	T-0012: A Trex employee has no firewall configuration review policy can increase network vulnerability, which leads to successful cyber attacks, resulting in reduced integrity and/or confidentiality of web server, AD Server, AD Database, sensitive customer information, user account information, network security, function of programs and network function T-0011: A Trex Employee never reviews user access rights can users having access to unnecessary systems, which leads to unauthorized configuration change, resulting in reduced integrity and/or availability of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token and function of programs T-0010: A Trex Employee does not get locked out of account after 3 unsuccessful login attempts can an attacker can use brute force password attacks, which leads to unwarranted access to the network, resulting in reduced integrity and/or confidentiality of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token, network security and function of programs T-0009: A Trex Employee Does not take required cyber security training can leave employees vulnerable to cyber attacks, which leads to successful phishing attacks, resulting in reduced integrity, availability and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account information, network security, OKTA Token, function of programs and network function T-0008: An internal actor Does not block incoming traffic to the Azure Firewall can allow external access to the network, which leads to degraded network security, resulting in reduced availability, integrity and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0007: An external threat actor bypasses azure firewall can access the network unknowingly, which leads to a network breach, resulting in reduced integrity, confidentiality and/or availability of web server, AD, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0006: An external threat actor Firewall rules are misconfigured, or there is an alternate route into the VNet that bypasses the Azure Firewall/NAT Gateway. can Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in the Application Plane, which leads to Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in the Application Plane, resulting in reduced availability, integrity and/or confidentiality of web server, AD, AD Server, AD Database and

Assets Number	Asset	Related Threats
		network function T-0003: An external threat actor has external access to TCP port 22 can Brute force attack, which leads to remote access to the attacker, resulting in reduced integrity, confidentiality and/or availability of network function, user account, web server, AD Server and sensitive customer information T-0002: An external threat actor traffic to port 53 is open can allow external access to the network, which leads to DDOS attacks are able to be performed, resulting in reduced integrity and/or availability of network function, web server, AD Database and AD Server T-0001: An internal actor Internal admin made an unauthorized change to the configuration of the active directory to not force MFA access for privileged users can bypass MFA, which leads to all users in AD permission would be changed, resulting in reduced confidentiality of AD, AD Server, AD Database and OKTA Token
AS-0008	AD Database	T-0012: A Trex employee has no firewall configuration review policy can increase network vulnerability, which leads to successful cyber attacks, resulting in reduced integrity and/or confidentiality of web server, AD Server, AD Database, sensitive customer information, user account information, network security, function of programs and network function T-0011: A Trex Employee never reviews user access rights can users having access to unnecessary systems, which leads to unauthorized configuration change, resulting in reduced integrity and/or availability of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token and function of programs T-0010: A Trex Employee does not get locked out of account after 3 unsuccessful login attempts can an attacker can use brute force password attacks, which leads to unwarranted access to the network, resulting in reduced integrity and/or confidentiality of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token, network security and function of programs T-0009: A Trex Employee Does not take required cyber security training can leave employees vulnerable to cyber attacks, which leads to successful phishing attacks, resulting in reduced integrity, availability and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account information, network security, OKTA Token, function of programs and network function T-0008: An internal actor Does not block incoming traffic to the Azure Firewall can allow external access to the network, which leads to degraded network security, resulting in reduced availability, integrity and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0007: An external threat actor bypasses azure firewall can access the network unknowingly, which leads to a network breach, resulting in reduced integrity, confidentiality and/or availability of web server, AD, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0006: An external threat actor Firewall rules are misconfigured, or there is an alternate route into the VNet that bypasses the Azure Firewall/NAT Gateway. can Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in the Application Plane, which leads to Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in the Application Plane, resulting in reduced availability, integrity and/or confidentiality of web server, AD, AD Server, AD Database and network function T-0002: An external threat actor traffic to port 53 is open can allow external access to

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		the network, which leads to DDOS attacks are able to be performed, resulting in reduced integrity and/or availability of network function, web server, AD Database and AD Server T-0001: An internal actor Internal admin made an unauthorized change to the configuration of the active directory to not force MFA access for privileged users can bypass MFA, which leads to all users in AD permission would be changed, resulting in reduced confidentiality of AD, AD Server, AD Database and OKTA Token
AS-0009	network function	T-0012: A Trex employee has no firewall configuration review policy can increase network vulnerability, which leads to successful cyber attacks, resulting in reduced integrity and/or confidentiality of web server, AD Server, AD Database, sensitive customer information, user account information, network security, function of programs and network function T-0009: A Trex Employee Does not take required cyber security training can leave employees vulnerable to cyber attacks, which leads to successful phishing attacks, resulting in reduced integrity, availability and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account information, network security, OKTA Token, function of programs and network function T-0006: An external threat actor Firewall rules are misconfigured, or there is an alternate route into the VNet that bypasses the Azure Firewall/NAT Gateway. can Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in the Application Plane, which leads to Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in the Application Plane, resulting in reduced availability, integrity and/or confidentiality of web server, AD, AD Server, AD Database and network function T-0005: An internal actor does not keep SIEM programs up to date can cause potential network breaches to be missed, which leads to lower overall network security, resulting in reduced integrity of sensitive customer information, network function, user account, web server, AD, network security and function of programs T-0003: An external threat actor has external access to TCP port 22 can Brute force attack, which leads to remote access to the attacker, resulting in reduced integrity, confidentiality and/or availability of network function, user account, web server, AD Server and sensitive customer information T-0002: An external threat actor traffic to port 53 is open can allow external access to the network, which leads to DDOS attacks are able to be performed, resulting in reduced integrity and/or availability of network function, web server, AD Database and AD Server
AS-0010	OKTA Token	T-0011: A Trex Employee never reviews user access rights can users having access to unnecessary systems, which leads to unauthorized configuration change, resulting in reduced integrity and/or availability of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token and function of programs T-0010: A Trex Employee does not get locked out of account after 3 unsuccessful login attempts can an attacker can use brute force password attacks, which leads to unwarranted access to the network, resulting in reduced integrity and/or confidentiality of AD Server, web server, AD Database, sensitive customer information, user account information, OKTA Token, network security and function of programs T-0009: A Trex Employee Does not take required cyber security training can leave employees vulnerable to cyber attacks, which leads to successful phishing attacks, resulting in reduced integrity, availability and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account information,

Assets Number	Asset	Related Threats
		network security, OKTA Token, function of programs and network function T-0008: An internal actor Does not block incoming traffic to the Azure Firewall can allow external access to the network, which leads to degraded network security, resulting in reduced availability, integrity and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0007: An external threat actor bypasses azure firewall can access the network unknowingly, which leads to a network breach, resulting in reduced integrity, confidentiality and/or availability of web server, AD, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0001: An internal actor Internal admin made an unauthorized change to the configuration of the active directory to not force MFA access for privileged users can bypass MFA, which leads to all users in AD permission would be changed, resulting in reduced confidentiality of AD, AD Server, AD Database and OKTA Token
AS-0011	AD	T-0009: A Trex Employee Does not take required cyber security training can leave employees vulnerable to cyber attacks, which leads to successful phishing attacks, resulting in reduced integrity, availability and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account information, network security, OKTA Token, function of programs and network function T-0008: An internal actor Does not block incoming traffic to the Azure Firewall can allow external access to the network, which leads to degraded network security, resulting in reduced availability, integrity and/or confidentiality of AD, web server, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0007: An external threat actor bypasses azure firewall can access the network unknowingly, which leads to a network breach, resulting in reduced integrity, confidentiality and/or availability of web server, AD, AD Server, AD Database, sensitive customer information, user account, user account information, network security, function of programs and OKTA Token T-0006: An external threat actor Firewall rules are misconfigured, or there is an alternate route into the VNet that bypasses the Azure Firewall/NAT Gateway. can Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in the Application Plane, which leads to Attackers discover and use the unprotected path (bypassing the firewall) to send malicious traffic directly to servers in the Application Plane, resulting in reduced availability, integrity and/or confidentiality of web server, AD, AD Server, AD Database and network function T-0005: An internal actor does not keep SIEM programs up to date can cause potential network breaches to be missed, which leads to lower overall network security, resulting in reduced integrity of sensitive customer information, network function, user account, web server, AD, network security and function of programs T-0001: An internal actor Internal admin made an unauthorized change to the configuration of the active directory to not force MFA access for privileged users can bypass MFA, which leads to all users in AD permission would be changed, resulting in reduced confidentiality of AD, AD Server, AD Database and OKTA Token